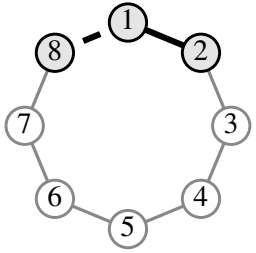
Positions on the auxiliary cycle C_8

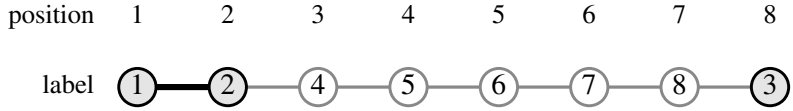
One compatible labeling and its spanning path

Shading: positions assigned to S (left), vertex labels in $S = \{1, 2, 3\}$ (right)

(a) Positive weight



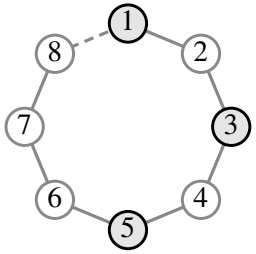
$A = \{1, 2, 8\}, \quad k(A) = 2, \quad w_A = 1/8$



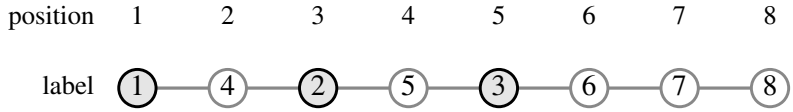
Weight of this ordering: $\nu_\sigma = w_A/(3!\,5!) = 1/5760$

Edges of this path inside S: 1

(b) Negative weight



$A = \{1, 3, 5\}, \quad k(A) = 0, \quad w_A = -1/16$



Weight of this ordering: $\nu_\sigma = w_A/(3!\,5!) = -1/11520$

Edges of this path inside S: 0